SAMPLE

History of Astronomy: Gravitational Waves

A New Window on the Universe

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Introduction

In the middle of the second decade of the twenty-first century, astronomy gained a new sense. For all of its previous history, the science had read the sky through light, every observation from the first dawn watch to the great space telescopes being a reading of electromagnetic radiation; then, in a single announcement, a new channel opened, and the universe began to be heard as well as seen. The new channel was gravitational waves, ripples in the fabric of space and time itself, and their detection was the end of a hundred-year story that begins with a paper of pure theory and ends with instruments that can feel the shudder of two black holes colliding a billion light-years away.

This volume tells that story as a piece of the history of astronomy, which is the series it belongs to, and the telling has a shape the reader should know at the start: a century of waiting, in which gravitational waves were predicted, doubted, re-derived, and hunted for half a century without success; then a decade of confirmation, in which indirect evidence from a famous pulsar made the doubters quiet and the detectors went from dream to steel; then a moment, and after the moment, a new astronomy, in which the detections arrived, the catalogs grew, and the sky opened a second time. It is a story with everything a history of science can want: a theory ahead of its instruments, a prediction that took a hundred years to verify, a Nobel-worthy indirect detection, an engineering project of almost absurd ambition, and a first event so violent that for a fraction of a second it outshone, in pure power, every star in the observable universe together.

The subject rewards a careful reader because it is deep physics wearing intuitive clothes. A gravitational wave is simple to say and strange to hold in the mind: space itself stretches and squeezes as the wave passes, so that the distance between two freely floating objects changes slightly, rhythmically, without either object moving through space at all. The chapters ahead build the idea honestly, from the theory that produced it, through the characteristics that make it both precious and almost impossible to detect, to the engineering miracle of the interferometers that finally caught it, and the reader needs no mathematics beyond the few numbers that appear in words.

A note on the division of labor among books: this volume is the history of the waves and their science, and it stays close to that subject. The great ground-based observatory that made the detection has its own companion volume in this publishing family, where the instrument's own story, its corridors, its optics, its upgrades, is told in full; the present book covers the detector only as the waves' history requires, as the instrument that ended the century of waiting. The same courtesy is extended to the neighboring subjects, the binary pulsar, the neutron-star merger, the instruments of the future, each treated here as the wave story needs them, and in full where they belong. What follows is the biography of an idea, from a page of theory to a new sense, and the sky has not been the same since.

A paragraph on the reader's stake in the subject, because a century of physics can feel distant from a life. The reader holds in their body the answer to what the waves do: nothing, at any strength a human will ever stand near, because the stretch of space is gentle even at its most violent source, and the passing wave would move the reader's head and feet by less than an atom's width, unnoticed by the nerves; yet the same gentle stretch, caught by the right machine, weighed two dead stars a billion light-years away. The gap between the imperceptible shiver and the weighed catastrophe is the whole subject, and the reader who follows the book will cross the gap in the reverse direction of the founders, starting at the discovery everyone knows and walking back to the theory no one believed, arriving, at the end, with the founders' own view: that the universe is stranger than it looks, and that looking longer is how the strangeness is read.

Chapter 1: The Discovery of Gravitational Waves

The discovery of gravitational waves is really three discoveries at seventy-year intervals, and the first one was made with pencil and paper. The theory of general relativity was published in 1915, and within a year its author had taken the new equations of curved space and time and found that they admitted a wave-like solution: a disturbance in the geometry of space could propagate at the speed of light, carrying energy away from its source. The prediction was almost accidental, a byproduct of the theory's structure, and it came surrounded by doubt: the waves looked, at first, like a mathematical artifact, a coordinate effect rather than a physical thing, and even the theory's own author spent decades unsure whether they were real, at one point publishing a paper arguing they did not exist. The first chapter of the wave story is therefore a chapter of controversy, in which the theory predicted and its own community hesitated.

The second discovery was the revival, and it belongs to the middle of the twentieth century. The mathematical dust settled: the waves were real, they carried energy, and they were produced, above all, by accelerating masses in the quadrupole pattern that only asymmetric motion can produce, so that binaries of compact stars were the sky's great wave sources, inspiraling and shrinking as the waves drained their orbital energy. The revival had a physical demonstration at its heart, the famous sticky bead argument that settled the energy question in the fifties: a wave passing through a row of beads on a rod would jiggle them, friction would warm the rod, and the heat had to come from somewhere, so the waves carried real energy. With the physics established, the hunt began, and the hunt's first act was a false start that deserves its honest paragraph below: Joseph Weber's resonant aluminum bars, claimed in the sixties to detect waves daily, could not be replicated, and the field learned, expensively, what a detection claim must survive.

The third discovery was the one the world noticed. In the last quarter of the twentieth century, a binary pulsar, two neutron stars orbiting each other in a tight dance, was found to be losing orbital energy at exactly the rate the wave theory demanded, an indirect detection that made the subject respectable and won its discoverers the Nobel prize; and the response to that respectability was the boldest instrument project in the history of astronomy, kilometer-scale laser interferometers built to catch the wave directly. The instrument worked. In the first weeks of its first observing run, the signature arrived, the merger of two black holes, the chirp rising through the band, the final coalescence, the ringdown, the whole event over in a fifth of a second, and the century of waiting ended on a September morning in the announcement heard round the world. The discovery chapter closes with the date every textbook now carries: prediction, 1916; indirect evidence, 1974; direct detection, 2015; and the sky, from that day, audible.

A paragraph on the sociology of the century, because the wave story's human texture is part of its history. The theory's early decades saw the waves dismissed in reviews, argued in correspondence, and rescued by a handful of physicists whose names the field now honors; the postwar revival was driven by a generation that included experimenters who spent careers on instruments their colleagues doubted, and the field's culture was forged in that doubt, a discipline of patience, of skepticism trained on its own claims, and of institutions built to outlast any single scientist's career. The sociology matters to the history because it explains the discovery's texture when it finally came: a field that had learned not to announce until the checks were done, and an announcement, when it came, whose care was itself the century's product, the doubt institutionalized as method.

A paragraph on the prediction's unease, the waves their own author half-doubted. The file's subject entered physics sideways: not as an observation, not even as a purpose, but as a logical consequence, the new theory of gravity yielding, in its mathematics, an unavoidable by-product, ripples running through space and time themselves radiated by accelerating masses; yet the consequence embarrassed even its author, who worried that the waves might be coordinate fictions, artifacts of the mathematics with no physical existence, the waves spending their early decades as a disputed implication, believed by some, doubted by others, and doubted, at intervals, by the very mind that produced them. The opening note worth recording: the subject's early history is a caution against the assumption that consequences are discoveries, the waves having existed on paper for forty years before anyone knew whether the paper was describing the world, the file's later chapters, the detectors and the detections, being the long-delayed audit of a by-product, the ripple finally promoted from artifact of theory to fact of nature by machines that took the dispute seriously when the theorists had grown tired of it.

A second opening card, on the famous detour, the retraction that had to be un-retracted. The subject's oddest early episode deserves the file's telling: in his later years, working with an assistant on a related problem, the theory's own author briefly convinced himself that the waves were indeed fictions, deriving a proof of their non-existence and submitting the retraction for publication; the journal's editor, a patient man, declined to publish the refutation, and history preserved the irony, the world's most famous physicist talked out of disbelieving his own theory's waves by the peer review he disdained, the retraction never printed, the waves surviving their author's second thoughts. The closing card for the opening: the episode is the file's best story of the self-correcting machinery of science working in reverse, the referee's polite refusal having protected a truth the author had abandoned, and the reader should hold the tale for its honest lesson, that even the founders' convictions carry no privilege, the waves having needed, to settle their existence, not authority but measurement, the retraction-un-retraction standing as the subject's strange prelude, the truth kept alive by an editor's returned envelope.